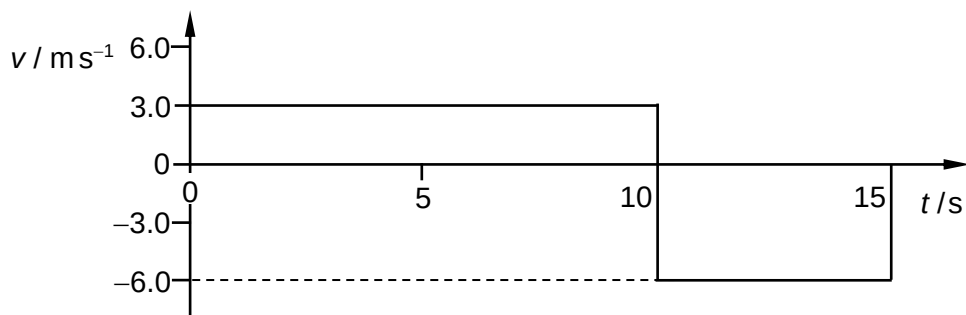


- 5 A radio-controlled toy car travels along a straight line for a time of 15 s.

The variation with time  $t$  of the velocity  $v$  of the car is shown below.



What is the average velocity of the toy car for the 15 s journey?

- A**  $-1.5 \text{ m s}^{-1}$     **B**  $0.0 \text{ m s}^{-1}$     **C**  $4.0 \text{ m s}^{-1}$     **D**  $4.5 \text{ m s}^{-1}$

Area under the graph =  $\Delta s$  = zero

$$v_{\text{ave}} = \Delta s / t = 0$$

6

**Answer: B**

6 A tennis ball is thrown horizontally in air from the top of a tall building. The effect of air resistance